

Unit 1: Data Communication Fundamentals & Physical Layer

Handwritten Student Notebook

Syllabus Topics Covered:

OFFICIAL SYLLABUS TOPICS COVERED Data Communication: 5 components, data representation, directional data flow (Simplex, Half-Duplex, Full-Duplex). Networks & Topologies: Network criteria (Performance, Reliability, Security), Mesh, Star, Bus, Ring, Hybrid. Formulas for links...

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Data Communication Fundamentals & Network Topologies

§ 1

Data communication is the exchange of data between two devices via some form of transmission medium (such as a wire cable or wireless radio link). For data communication to be effective, it depends on four fundamental characteristics: delivery, accuracy, timeliness, and jitter.

5 FIVE COMPONENTS OF DATA COMMUNICATION:

1. **Message:** The information (data) communicated (text, numbers, pictures, audio, video).
2. **Sender:** The device that creates and transmits the data message (workstation, server, phone, video camera).
3. **Receiver:** The device that receives the transmitted message (workstation, printer, TV monitor).
4. **Transmission Medium:** The physical physical conduit over which the message travels from sender to receiver (twisted-pair wire, coaxial cable, fiber-optic cable, or radio waves).
5. **Protocol:** A set of formal rules that govern data communication. It represents an agreement between communicating devices without which two devices may be physically connected but unable to communicate.

Data Flow Modes

Flow Mode	Directionality	Channel Capacity Utilization	Real-world Examples
Simplex	Strictly unidirectional. Only one device transmits; the other only receives.	100% of bandwidth dedicated to one direction.	Keyboard to CPU, traditional broadcast television, FM radio.
Half-Duplex	Bidirectional, but not at the same time . Each station can transmit and receive, but sequentially.	Entire capacity shared alternately.	Walkie-talkies, CB radio, classic Ethernet collision domain.

Flow Mode	Directionality	Channel Capacity Utilization	Real-world Examples
Full-Duplex	Simultaneously bidirectional. Both stations can transmit and receive concurrently.	Capacity divided between directions or carried over separate physical circuits.	Telephone network, modern switched full-duplex Ethernet.

Physical Topologies & Structural Formulas

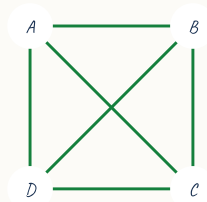
The **topology** of a network defines how nodes are connected geometrically to one another. Two or more devices connect to a link; two or more links form a topology.

Topology	Physical Links Formula	I/O Ports per Node	Advantages	Disadvantages
Mesh (Fully Connected)	$\frac{n(n-1)}{2}$	$n-1$	Dedicated links (no traffic congestion), robust (failure of one link affects only that link), privacy/security, easy fault isolation.	Massive cabling requirement, high cost of hardware and I/O ports, difficult installation.
Star	n (to central hub/switch)	1	Inexpensive, easy to install and reconfigure, robust (if one link fails, only that node goes down; other links stay active).	Single Point of Failure: If the central hub/switch crashes, the entire network is disabled.

Topology	Physical Links Formula	I/O Ports per Node	Advantages	Disadvantages
Bus	$\frac{1}{2}n(n-1)$ long backbone cable + n drop lines	1 (tap)	Minimal cabling, easy to install for small footprints.	Difficult fault isolation, signal reflection at ends requires terminators, degradation as nodes are added; break in backbone halts whole net.
Ring	n point-to-point links	2 (repeater in / out)	Deterministic access, easy to install, each repeater regenerates weak signals.	Unidirectional ring failure breaks entire network (unless dual counter-rotating ring like FDDI is used).

Comparison of Physical Network Topologies

Fully Connected Mesh (n=4)



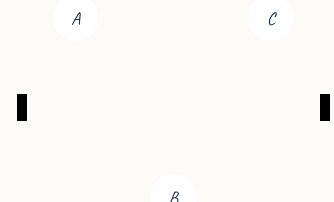
Links = $\frac{4(3)}{2} = 6$ links

Star Topology



Dedicated point-to-point to HUB

Bus Topology



Single shared backbone cable

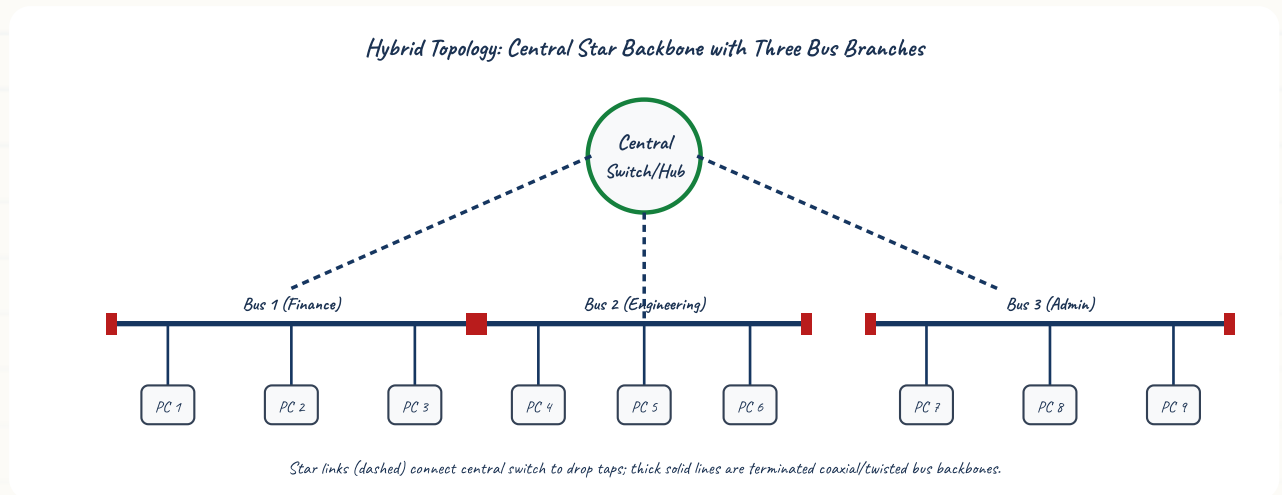


Figure 1.2: Hybrid Topology – Central Star Backbone with Three Bus Branches

As requested in Ramaiah CIE-1 Exam: combines star resilience with bus cabling efficiency for departmental workgroups.

QUICK RECALL

- 5 Components: Message, Sender, Receiver, Medium, Protocol.
- Mesh requires $\frac{n(n-1)}{2}$ links and $n - 1$ ports per node. Star requires n links to a central hub.
- Simplex = 1-way; Half-Duplex = 2-way alternate; Full-Duplex = 2-way concurrent.

The OSI 7-Layer Model vs TCP/IP Protocol Suite

§ 2

The **Open Systems Interconnection (OSI)** model, developed by the ISO in 1984, is a 7-layer theoretical reference framework. In contrast, the **TCP/IP Protocol Suite** is the pragmatic, 5-layer hierarchical protocol suite upon which the global Internet is built.

Layer #	Layer Name	Data Unit (PDU)	Core Responsibilities & Duties	Primary Protocols
Layer 7	Application	Message	Provides user interface and network services to applications. File transfers, directory services, web browsing, email access.	HTTP/HTTPS, DNS, SMTP, FTP, SSH
Layer 6	Presentation	Data	Translation (ASCII/EBCDIC), Encryption/Decryption (cryptographic security), and Compression (lossless/lossy).	SSL/TLS, JPEG, MPEG
Layer 5	Session	Data	Dialog control (half/full duplex), Synchronization (inserting checkpoints into data streams for recovery).	NetBIOS, RPC, PPTP
Layer 4	Transport	Segment / Datagram	Process-to-process delivery via port addressing, segmentation & reassembly, connection control, flow control, error control.	TCP, UDP, SCTP
Layer 3	Network	Packet / Datagram	Source-to-destination (host-to-host) delivery , logical IP addressing, routing (path selection), datagram fragmentation.	IPv4, IPv6, ICMP, OSPF, BGP
Layer 2	Data Link	Frame	Hop-to-hop (node-to-node) framing , physical MAC addressing, flow control (feedback), error detection (CRC) and correction.	Ethernet (802.3), Wi-Fi (802.11), PPP, ARP

Layer #	Layer Name	Data Unit (PDU)	Core Responsibilities & Duties	Primary Protocols
1	Physical	Bits	Transmission of raw unstructured bits over physical medium. Mechanical/electrical specifications, line coding, data rates, bit synchronization.	RJ-45, V.35, Cat6, Fiber optics

MNEMONIC FOR THE 7 LAYERS:

Top-down (Layer 7 → 1): "All People Seem To Need Data Processing"

Bottom-up (Layer 1 → 7): "Please Do Not Throw Sausage Pizza Away"

OSI vs TCP/IP Comparison

Feature / Dimension	OSI 7-Layer Reference Model	TCP/IP Protocol Suite
Number of Layers	7 Layers (Application, Presentation, Session, Transport, Network, Data Link, Physical)	4 or 5 Layers (Application [merges 7,6,5], Transport, Network/Internet, Data Link, Physical)
Development Approach	Theoretical concept first; protocols designed later to fit the layers.	Protocols developed first (TCP, IP); model was a description of existing protocols.
Transport Delivery	Strictly connection-oriented service at transport layer.	Both connection-oriented (TCP) and connectionless (UDP) services.
Network Layer Delivery	Supports both connectionless and connection-oriented network services.	Strictly connectionless datagram service at Internet layer (IP).
Commercial Dominance	Never widely deployed commercially; used primarily as a teaching reference.	The de-facto architecture and foundation of the worldwide Internet.

QUICK RECALL

- Physical = bits (raw signals); Data Link = frames (hop-to-hop / MAC); Network = packets (host-to-host / IP).
- Transport = segments (process-to-process / port); Application = user network access.
- Presentation handles Translation, Compression, Encryption; Session handles Dialog Control & Checkpoints.

Data and Signals (Time vs Frequency Domain)**§ 3**

Information must be transformed into electromagnetic signals before transmission across a physical medium. Signals can be continuous (analog) or discrete (digital).

CHARACTERISTICS OF A PERIODIC SINE WAVE:

A periodic analog sine wave is mathematically represented as:

$$s(t) = A \sin(2\pi ft + \phi)$$

- **Peak Amplitude (A):** The absolute value of its highest intensity, proportional to the energy carried by the signal (measured in Volts, Amperes, or Watts).
- **Period (T) and Frequency (f):** Period is the time taken to complete one full cycle (seconds). Frequency is the number of cycles per second (Hertz, Hz). They are inverse:

$$f = \frac{1}{T}, \quad T = \frac{1}{f}$$

- **Phase (ϕ):** The fractional position of the waveform relative to time zero (measured in degrees or radians).
- **Wavelength (λ):** The physical distance a signal propagates in one period through a medium with propagation speed c :

$$\lambda = \frac{c}{f} = c \times T$$

Time Domain vs Frequency Domain (Fourier Analysis)

Fourier Analysis proves that any composite periodic signal can be decomposed into an infinite series of simple sinusoidal waves with distinct amplitudes, frequencies, and phases:

Domain Representation	Horizontal Axis	Vertical Axis	Information Displayed
Time Domain Plot	Time (t)	Instantaneous Amplitude (v)	Shows signal voltage value at each instant of time (oscilloscope view).
Frequency Domain Plot	Frequency (f)	Peak Amplitude (A)	Shows only the frequency components and their peak intensities (spectrum analyzer view).

SIGNAL BANDWIDTH:

The **bandwidth** of a composite signal is the difference between the highest and lowest frequencies contained in that signal:

$$B = f_{\text{high}} - f_{\text{low}}$$

Example: If a composite periodic signal contains frequencies from 1000 Hz to 5000 Hz, its bandwidth is $B = 5000 - 1000 = 4000 \text{ Hz} = 4 \text{ kHz}$.

QUICK RECALL

- $f = 1/T$, $\lambda = c/f$. Phase ϕ is the angular offset at $t = 0$.
- Time-domain shows amplitude vs time; Frequency-domain shows peak amplitude vs frequency.
- Bandwidth $B = f_{\text{high}} - f_{\text{low}}$.

Transmission Impairments & Decibel Calculations

§ 4

As a signal travels through a physical transmission medium, it undergoes degradation caused by physical imperfections of the channel. The three primary transmission impairments are **attenuation**, **distortion**, and **noise**.

1

Attenuation (Loss of Energy): Signal power dissipates as heat due to the electrical resistance of the conductor. Over long distances, amplifiers or repeaters must regenerate the signal. Quantified using **Decibels (dB)**.

2

Distortion (Alteration of Shape): In composite signals, different frequency components travel at different propagation speeds through the medium (delay distortion) and experience different attenuation levels. They arrive at the receiver with shifted phases, altering the composite shape.

3

Noise (Extraneous Interference): Unwanted external energy coupled into the channel:

- *Thermal noise:* Random kinetic motion of electrons in conductors (white noise, Johnson-Nyquist noise).
- *Induced noise:* Electromagnetic interference (EMI) from motors, transformers, and appliances.
- *Crosstalk:* Inductive coupling between adjacent wire pairs.
- *Impulse noise:* Sudden voltage spikes from lightning or power surges.

Decibel (dB) Formulas

DECIBEL EQUATIONS:

For power levels P_1 (input) and P_2 (output):

$$dB = 10 \log_{10} \left(\frac{P_2}{P_1} \right)$$

- If $P_2 < P_1$, dB is negative (attenuation / loss).
- If $P_2 > P_1$, dB is positive (amplification / gain).
- When power is halved ($P_2/P_1 = 0.5$): $dB = 10 \log_{10}(0.5) \approx -3.010$ dB.

For voltages V_1 and V_2 across equal impedances:

$$dB = 20 \log_{10} \left(\frac{V_2}{V_1} \right)$$

Signal-to-Noise Ratio (SNR)

The **Signal-to-Noise Ratio** is the ratio of average signal power to average noise power:

$$\text{SNR} = \frac{P_{\text{signal}}}{P_{\text{noise}}}$$

$$\text{SNR}_{\text{dB}} = 10 \log_{10}(\text{SNR})$$

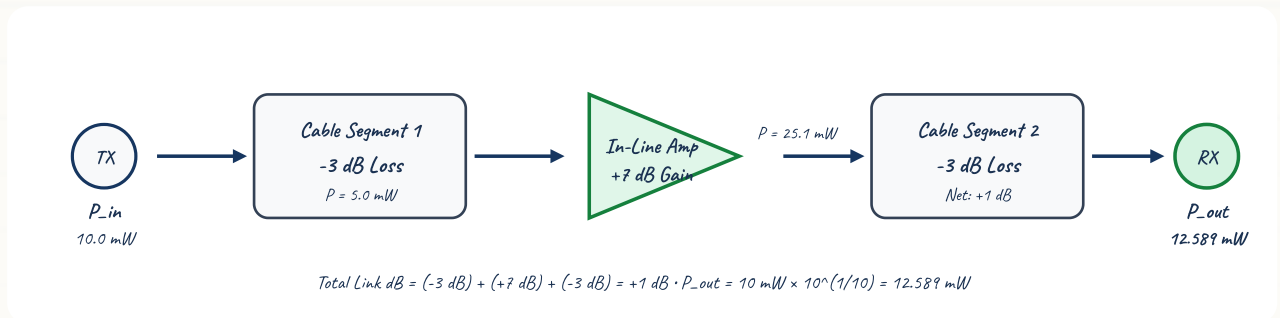


Figure 1.3: Signal Power Evolution across Cascaded Cable & Amplifier Stages

Net decibels are additive: $-3 \text{ dB} + 7 \text{ dB} - 3 \text{ dB} = +1 \text{ dB}$. Input 10 mW yields 12.589 mW output.

WORKED NUMERICAL 4.1 · EASY / FOUNDATION [3 MARKS]

Decibel Rating for Halved and Doubled Signal Power

Compute the decibel (dB) rating of a transmission link when:

1. The signal power drops to exactly half of its initial value ($P_2/P_1 = 0.5$).
2. The signal power is doubled by an active repeater ($P_2/P_1 = 2.0$).

Solution:

Case 1: Power Halved:

$$\text{dB} = 10 \log_{10} \left(\frac{P_2}{P_1} \right) = 10 \log_{10}(0.5) = 10 \times (-0.30103) \approx -3.01 \text{ dB}$$

Result: A 3 dB loss (-3.01 dB) represents a 50% drop in signal power

Case 2: Power Doubled:

$$\text{dB} = 10 \log_{10}(2.0) = 10 \times (+0.30103) \approx +3.010 \text{ dB}$$

Result: A 3 dB gain ($+3.01 \text{ dB}$) doubles the signal power

WORKED NUMERICAL 4.2 · MEDIUM / EXAM STANDARD [5 MARKS]**Signal Power through Cascaded Transmission Stages**

A signal with an input power of $P_{in} = 10 \text{ mW}$ passes through three cascaded transmission segments:

- Segment 1: Transmission cable with an attenuation of -3 dB
- Segment 2: In-line amplifier with a gain of $+7 \text{ dB}$
- Segment 3: Secondary cable section with an attenuation of -3 dB

Compute: (a) The net decibel rating of the link, and (b) The final output power P_{out} .

Part (a): Net decibel rating:

Decibels are additive across cascaded stages:

$$dB_{total} = dB_1 + dB_2 + dB_3 = (-3) + (+7) + (-3) = +1 \text{ dB}$$

Part (b): Output power P_{out} :

$$dB_{total} = 10 \log_{10} \left(\frac{P_{out}}{P_{in}} \right) \implies 1 = 10 \log_{10} \left(\frac{P_{out}}{10} \right)$$

$$\log_{10} \left(\frac{P_{out}}{10} \right) = \frac{1}{10} = 0.1 \implies \frac{P_{out}}{10} = 10^{0.1} \approx 1.2589$$

Final Output Power: $P_{out} = 10 \times 1.2589 = 12.589 \text{ mW}$ (+1 dB net gain)

WORKED NUMERICAL 4.3 · EXAM-LEVEL / SYSTEM DESIGN [6 MARKS]**Cable Loss over Distance and Repeater Spacing**

A coaxial cable exhibits an attenuation rate of 0.3 dB/km. An input signal with power $P_{in} = 200$ mW is injected at the transmitter.

1. Calculate the signal power at a distance of 10 km.
2. If the link extends to 100 km and repeaters offering 15 dB gain each are inserted, calculate the minimum number of repeaters required to fully compensate for cable attenuation.

Part 1: Signal power at 10 km:

$$\text{Total loss} = 10 \text{ km} \times 0.3 \text{ dB/km} = 3.0 \text{ dB}$$

$$\text{dB} = -3.0 = 10 \log_{10} \left(\frac{P_{out}}{200} \right) \implies \log_{10} \left(\frac{P_{out}}{200} \right) = -0.30$$

$$\frac{P_{out}}{200} = 10^{-0.30} \approx 0.50119$$

$$\text{Power at 10 km: } P_{out} = 200 \times 0.50119 = 100.24 \text{ mW}$$

Part 2: Repeaters required over 100 km:

$$\text{Total link loss over 100 km} = 100 \text{ km} \times 0.3 \text{ dB/km} = 30.0 \text{ dB}$$

$$\text{Number of 15 dB repeaters} = \frac{\text{Total Loss}}{\text{Gain per Repeater}} = \frac{30.0 \text{ dB}}{15.0 \text{ dB}} = 2$$

Repeaters Required: Exactly 2 in-line amplifiers (spaced at 33.3 km and 66.7 km)

QUICK RECALL

- Impairments: Attenuation (loss), Distortion (phase shift/delay), Noise (external interference).
- $\text{dB} = 10 \log_{10}(P_2/P_1)$. Halving power = -3 dB; doubling power = $+3$ dB.
- $\text{SNR}_{dB} = 10 \log_{10}(\text{SNR})$. Linear $\text{SNR} = 10^{(\text{SNR}_{dB}/10)}$.

Data Rate Limits: Nyquist Bit Rate & Shannon Capacity

A central problem in data communications is determining the maximum theoretical data rate (in bits per second) that can be reliably transmitted over a transmission channel of bandwidth B . Two fundamental physical laws dictate these limits: **Nyquist's theorem** for noiseless channels and **Shannon's theorem** for noisy channels.

1. Nyquist Bit Rate (Noiseless Channel)

In 1928, Harry Nyquist showed that for an ideal, noiseless channel of bandwidth B , the maximum transmission rate depends on the number of discrete signal voltage levels L :

NYQUIST BIT RATE FORMULA:

$$C_{\text{Nyquist}} = 2 \times B \times \log_2(L) \quad [\text{bps}]$$

Where:

- B : Bandwidth of the channel in Hertz (Hz).
- L : Number of discrete signal levels used to represent data ($L \geq 2$).
- $\log_2(L)$: Number of bits carried per signal element ($r = \log_2 L$).

2. Shannon Capacity (Noisy Channel)

In 1948, Claude Shannon extended this to realistic channels subject to random thermal noise. The Shannon capacity represents the absolute theoretical upper bound for error-free communication:

SHANNON CHANNEL CAPACITY FORMULA:

$$C_{\text{Shannon}} = B \times \log_2(1 + \text{SNR}) \quad [\text{bps}]$$

Where:

- B : Channel bandwidth in Hertz (Hz).
- SNR: **Linear** Signal-to-Noise Ratio (NOT in decibels!).
- Note: As $\text{SNR} \rightarrow 0$ (extreme noise), $C \rightarrow 0$. Capacity cannot be increased by adding signal levels alone if constrained by noise.

WORKED NUMERICAL 5.1 · STANDARD EXAM PROBLEM [5 MARKS]**Shannon Capacity of a Noisy Telephone Channel**

A standard analog telephone line has a bandwidth of $B = 3000$ Hz (3 kHz) and a Signal-to-Noise Ratio of $\text{SNR}_{\text{dB}} = 35$ dB. Compute the theoretical channel capacity.

Step 1: Convert SNR_{dB} to linear SNR:

$$\text{SNR}_{\text{dB}} = 10 \log_{10}(\text{SNR}) \implies 35 = 10 \log_{10}(\text{SNR}) \implies \log_{10}(\text{SNR}) = 3.5$$

$$\text{SNR} = 10^{3.5} \approx 3162.28$$

Step 2: Apply Shannon Capacity Formula:

$$C = B \log_2(1 + \text{SNR}) = 3000 \times \log_2(1 + 3162.28) = 3000 \times \log_2(3163.28)$$

$$\log_2(3163.28) = \frac{\ln(3163.28)}{\ln(2)} \approx 11.6272$$

$$\text{Shannon Capacity: } C = 3000 \times 11.6272 = 34,881.6 \text{ bps} \approx 34.88 \text{ kbps}$$

Exam Comparison: At $\text{SNR}_{\text{dB}} = 30$ dB ($\text{SNR} = 1000$), $C = 3000 \times \log_2(1001) = 29.902$ kbps. The 35 dB rating yields 34.88 kbps.

WORKED NUMERICAL 5.2 · COMBINED DESIGN PROBLEM [8 MARKS · HIGH OCCURRENCE]**Signal Levels Required to Attain Shannon Limit**

A channel has a bandwidth of $B = 1$ MHz and a linear $\text{SNR} = 63$.

1. Find the Shannon theoretical capacity of the channel.
2. How many discrete signal voltage levels L must the transmitter employ to achieve this data rate under Nyquist's theorem?

Part 1: Shannon Capacity:

$$C = B \log_2(1 + \text{SNR}) = 10^6 \times \log_2(1 + 63) = 10^6 \times \log_2(64)$$

Since $64 = 2^6$, $\log_2(64) = 6$:

Shannon Capacity: $C = 10^6 \times 6 = 6.000$ Mbps

Part 2: Finding Required Signal Levels L :

Equate the Nyquist formula to the Shannon limit:

$$C_{\text{Nyquist}} = 2B \log_2(L) \implies 6 \times 10^6 = 2 \times (10^6) \times \log_2(L)$$

$$\log_2(L) = \frac{6 \times 10^6}{2 \times 10^6} = 3 \implies L = 2^3 = 8 \text{ levels}$$

Signal Levels: $L = 8$ discrete voltage levels (3 bits/signal element)

WORKED NUMERICAL 5.3 · RAMAIAH CIE-1 (2025) SOLVED [4 MARKS · QUESTION 1(a)]

Channel with Reduced Bit Rate below Shannon Limit

(Past Paper Question 1.a): "We have a channel with a 2-MHz bandwidth. The SNR for this channel is 63. What are the appropriate bit rate and signal level? (To calculate signal level, consider the bit rate to be 4 Mbps less than the upper limit)."

Step 1: Compute the Shannon Upper Limit:

$$B = 2 \text{ MHz} = 2 \times 10^6 \text{ Hz}, \quad \text{SNR} = 63$$

$$C_{\text{upper}} = B \log_2(1 + \text{SNR}) = 2 \times 10^6 \times \log_2(1 + 63) = 2 \times 10^6 \times 1$$

Step 2: Determine Specified Operating Bit Rate:

$$R = C_{\text{upper}} - 4 \text{ Mbps} = 12 \text{ Mbps} - 4 \text{ Mbps} = 8 \text{ Mbps}$$

Step 3: Determine Signal Level L using Nyquist formula:

$$R = 2B \log_2(L) \implies 8 \times 10^6 = 2 \times (2 \times 10^6) \times \log_2(L) = 4 \times 10^6 \log_2(L)$$

$$\log_2(L) = \frac{8 \times 10^6}{4 \times 10^6} = 2 \implies L = 2^2 = 4 \text{ levels}$$

CIE-1 Solution: Bit Rate $R = 8 \text{ Mbps}$ · Signal Level $L = 4 \text{ levels}$

3. Network Performance: Latency & Bandwidth-Delay Product (BDP)

*In addition to data rate limits, two metrics govern network performance: **latency (delay)** and the **Bandwidth-Delay Product**.*

LATENCY COMPONENTS:

The total latency (one-way delay) for a packet traversing a transmission link is the sum of four distinct delays:

$$\text{Total Latency} = T_{trans} + T_{prop} + T_{queue} + T_{proc}$$

- **Transmission Delay (T_{trans}):** Time needed to push all packet bits onto the wire:

$$T_{trans} = \frac{\text{Packet Length } (L) \text{ [bits]}}{\text{Transmission Rate } (R) \text{ [bps]}}$$

- **Propagation Delay (T_{prop}):** Time for a single bit to travel from sender to receiver:

$$T_{prop} = \frac{\text{Distance } (d) \text{ [m]}}{\text{Propagation Speed } (v) \text{ [m/s]}}$$

Typical speeds: $v \approx 2 \times 10^8$ m/s in copper/fiber; $v \approx 3 \times 10^8$ m/s in air/vacuum.

- **Queuing Delay (T_{queue}):** Time the packet waits in intermediate router buffers before transmission.
- **Processing Delay (T_{proc}):** Time required for routers to examine packet headers, verify checksums, and determine outgoing interface.

Packet Latency Component Breakdown (1000 km Fiber Link)

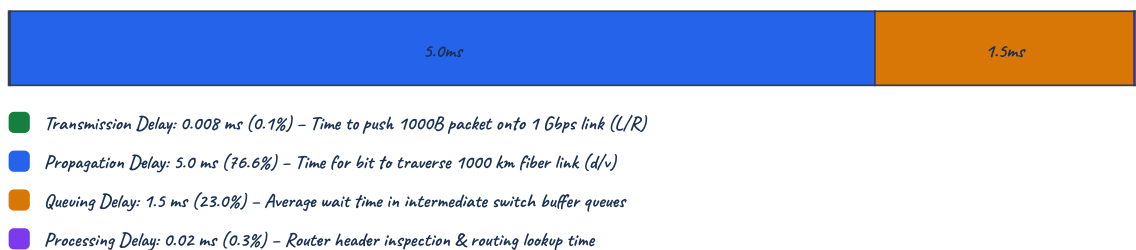


Figure 1.4: Packet Latency Component Breakdown for a 1000 km Optical Fiber Link

On high-speed links, propagation delay (5 ms) and queuing dominate over packet transmission time (8 μ s).

BANDWIDTH-DELAY PRODUCT (BDP):

The **Bandwidth-Delay Product (BDP)** defines the maximum number of bits that can fill the transmission link at any given instant ("bits in flight"):

$$\text{BDP} = \text{Bandwidth (R)} \times \text{Propagation Delay (T}_{\text{prop}})$$

Conceptually, envision the link as an empty water pipe connecting sender and receiver. The cross-sectional area of the pipe is the bandwidth (R), and its length is the propagation delay (T_{prop}). The volume of the pipe represents the BDP.

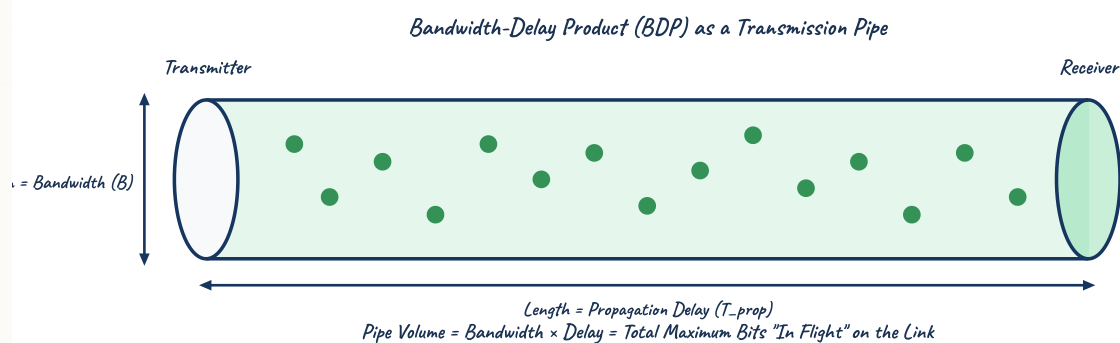


Figure 1.5: Conceptual Representation of Bandwidth-Delay Product as a Pipe

Volume of pipe = $R \times T_{\text{prop}}$. Critical for sizing sliding window buffers in TCP and ARQ protocols.

WORKED NUMERICAL 5.4 · STORE-AND-FORWARD MULTI-HOP LATENCY [5 MARKS]

Store-and-Forward Packet Transmission across Multiple Routers

A host transmits a 1000-Byte (8000-bit) packet to a destination host across $N = 3$ identical links separated by 2 intermediate store-and-forward routers. Each link has a bandwidth of $R = 10$ Mbps, distance $d = 100$ km, and propagation speed $v = 2 \times 10^8$ m/s. Each router has a processing delay of $100 \mu\text{s}$. Queuing delay is negligible (0). Calculate total end-to-end latency.

Step 1: Transmission delay per hop (T_{trans}):

$$T_{trans} = \frac{8000 \text{ bits}}{10 \times 10^6 \text{ bps}} = 0.0008 \text{ s} = 0.800 \text{ ms} = 800 \mu\text{s}$$

Step 2: Propagation delay per hop (T_{prop}):

$$T_{prop} = \frac{100 \times 10^3 \text{ m}}{2 \times 10^8 \text{ m/s}} = 0.0005 \text{ s} = 0.500 \text{ ms} = 500 \mu\text{s}$$

Step 3: Total End-to-End Latency:

In store-and-forward switching, each router must completely receive the packet before forwarding it. Thus, the packet is transmitted 3 times, propagates across 3 links, and is processed at 2 routers:

$$T_{total} = 3 \times T_{trans} + 3 \times T_{prop} + 2 \times T_{proc}$$

$$T_{total} = 3 \times (0.8 \text{ ms}) + 3 \times (0.5 \text{ ms}) + 2 \times (0.1 \text{ ms}) = 2.4 + 1.5 +$$

Total Multi-Hop Latency: $T_{total} = 4.100 \text{ ms}$ (4100 μs)

WORKED NUMERICAL 5.5 · GEO SATELLITE BDP & PROTOCOL EFFICIENCY [6 MARKS]

Bandwidth-Delay Product and Efficiency of Satellite Link

A geostationary earth orbit (GEO) satellite communication link operates at a data rate of $R = 10$ Mbps. The one-way propagation delay is $T_{prop} = 250$ ms (0.25 s).

1. Calculate the Bandwidth-Delay Product in bits and in Kilobytes.
2. If a Stop-and-Wait protocol is used with a frame size of $L = 1500$ Bytes (12,000 bits) and negligible ACK transmission time, calculate the channel efficiency (utilization η).

Part 1: Bandwidth-Delay Product:

$$\text{BDP} = R \times T_{prop} = (10 \times 10^6 \text{ bps}) \times 0.25 \text{ s} = 2,500,000 \text{ bits} = 312,500 \text{ Bytes} \approx 312.5 \text{ KB}$$

Satellite BDP: 2,500,000 bits (312.5 KB in flight)

Part 2: Stop-and-Wait Channel Efficiency:

$$T_{trans} = \frac{12,000 \text{ bits}}{10 \times 10^6 \text{ bps}} = 0.0012 \text{ s} = 1.2 \text{ ms}$$

$$\text{Round Trip Time (RTT)} = 2 \times T_{prop} = 2 \times 250 \text{ ms} = 500 \text{ ms}$$

$$\text{Total cycle time} = T_{trans} + \text{RTT} = 1.2 \text{ ms} + 500 \text{ ms} = 501.2 \text{ ms}$$

$$\eta = \frac{T_{trans}}{T_{trans} + 2T_{prop}} = \frac{1.2}{501.2} \approx 0.00239 = \mathbf{0.24\%}$$

Channel Utilization: $\eta = 0.24\%$ (Extremely wasteful; explains why sliding windows are mandatory)



Interactive Tool Available Online: ⚡ Interactive Latency & Bandwidth-Delay Explorer. Explore the step-by-step calculator and visualizer on the live site notes page.

QUICK RECALL

- Nyquist (noiseless): $C = 2B \log_2(L)$ bps.
- Shannon (noisy): $C = B \log_2(1 + \text{SNR})$ bps. Always convert SNR_{dB} to linear first!
- Shannon gives the upper theoretical ceiling; Nyquist tells how many levels L are required.

Digital Line Coding Schemes & Waveforms

§ 6

Line coding is the process of converting a digital sequence of binary 1s and 0s into a continuous digital signal for physical transmission. A well-designed line coding scheme must prevent DC components, avoid baseline wandering, and support self-synchronization.

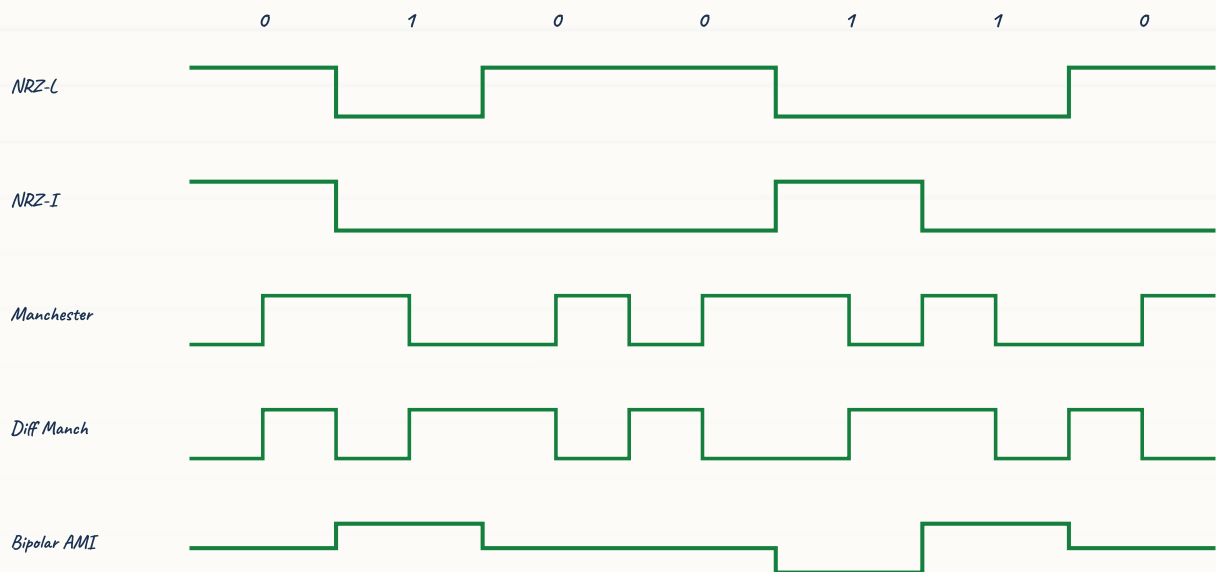
Core Properties of Line Coding

- **DC Component:** When voltage levels remain constant over time, a direct current (DC) component is created. DC components cannot pass through transformers or AC-coupled circuits and generate heat. A good line coding scheme should have zero DC component.
- **Baseline Wandering:** The receiver averages the received signal power to establish a threshold (baseline) for distinguishing 1s from 0s. Long strings of 0s or 1s shift this baseline, causing decoding errors.
- **Self-Synchronization:** Receiver and sender clocks must tick at the same rate. Transitions in the middle of bit intervals provide clock synchronization edges.

Taxonomy of Line Coding Schemes

Coding Family	Scheme Name	Encoding Rule	Self-Sync & DC Status
Unipolar	NRZ (Non-Return-to-Zero)	Positive voltage for binary 1; Zero voltage for binary 0.	Poor sync; large DC component; baseline wander.
	NRZ-L (Level)	Positive voltage represents binary 0; Negative voltage represents binary 1 (or vice versa).	No self-sync on consecutive identical bits; DC component present.
	NRZ-I (Invert)	Transition at the beginning of the bit interval represents binary 1; No transition represents binary 0.	Sync on 1s, but fails on consecutive 0s. Baseline wander on long 0s.
Polar	Manchester	Transition at the center of every bit interval: High-to-Low represents binary 1; Low-to-High represents binary 0 (IEEE 802.3 standard).	Perfect self-sync on every bit; Zero DC component ; requires $2\times$ bandwidth.
	Differential Manchester	Transition at center always occurs (clocking). Transition at beginning represents binary 0; No transition at beginning represents binary 1.	Perfect self-sync; noise-immune; widely used in Token Ring (802.5).
	AMI (Alternate Mark Inversion)	Zero voltage represents binary 0. Binary 1s alternate between $+V$ and $-V$.	Zero DC component ; but loses sync on long runs of 0s.
Bipolar	Pseudoternary	Inverse of AMI: Zero voltage represents binary 1; binary 0s alternate between $+V$ and $-V$.	Zero DC component; loses sync on long runs of 1s.

Line Coding Waveforms for Bit Stream: 0 1 0 0 1 1 0



QUICK RECALL

- Manchester: Mid-bit transition on every bit. 0 = Low-to-High; 1 = High-to-Low. Zero DC.
- Differential Manchester: Center transition always occurs. 0 = transition at start; 1 = no transition at start.
- AMI: 0 = Zero volts; 1s alternate between $+V$ and $-V$. Eliminates DC component.

One-Page Rapid Revision Sheet

★ REVISION

Topologies & Formulas

- Mesh: $\frac{n(n-1)}{2}$ links, $n - 1$ ports per node.
- Star: n links to central hub. Hub is single point of failure.
- Bus: 1 shared backbone cable + terminators.
- Ring: n repeaters; token passing.

OSI 7-Layer Summary

- Physical: raw bits, line coding, electrical specs.
- Data Link: frames, hop-to-hop delivery, MAC, CRC.
- Network: packets, host-to-host delivery, IP, routing.
- Transport: segments, process-to-process, ports, flow/error.
- Session/Pres/App: Dialog, Encryption/Compression, User services.

Impairments & Decibels

- Attenuation: energy loss in decibels.
- $dB = 10 \log_{10}(P_2/P_1)$. Halving power = -3 dB.
- Decibels are additive in cascaded stages ($dB_{tot} = \sum dB_i$).
- $SNR_{dB} = 10 \log_{10}(SNR)$.

Nyquist vs Shannon

- Nyquist: $C = 2B \log_2(L)$ bps (noiseless channel).
- Shannon: $C = B \log_2(1 + SNR)$ bps (noisy channel).
- Line Coding: Manchester has zero DC and mid-bit clock transition. AMI alternates $+V / -V$ for 1s.

Important Examination Questions

? QUESTIONS

2-Mark Questions (Short Answer)

Practice Problem:

2M State the Nyquist bit rate formula and define all symbols.

Solution:

Answer: $C = 2B \log_2(L)$ bps, where B is channel bandwidth in Hertz, L is the number of discrete signal voltage levels, and $\log_2 L$ is the number of bits carried per signal element.

Practice Problem:

2M What is the significance of the mid-bit transition in Manchester encoding?

Solution:

Answer: The mid-bit transition serves two crucial purposes: (1) It guarantees self-synchronization by providing a clocking edge on every single bit, and (2) It ensures equal positive and negative voltage areas, achieving a zero DC component.

Practice Problem:

2M Calculate the attenuation in dB when signal power is reduced to one-half.

Solution:

Answer: $dB = 10 \log_{10}(P_2/P_1) = 10 \log_{10}(0.5) = 10 \times (-0.30103) = -3.010$ dB. (Power loss of ~ 3 dB).

5-Mark Questions (Medium Length)

Practice Problem:

SM Compare the OSI 7-layer reference model with the TCP/IP protocol suite.

Solution:

Answer:

- **Layer count:** OSI has 7 layers; TCP/IP has 4 or 5 layers (combining Application, Presentation, and Session into one Application layer).
- **Design:** OSI was theoretical with protocols defined later; TCP/IP protocols were implemented first and model described them.
- **Network layer:** OSI supports both connectionless and connection-oriented services; TCP/IP provides strictly connectionless IP datagrams.
- **Transport layer:** OSI is strictly connection-oriented; TCP/IP supports both TCP (connection-oriented) and UDP (connectionless).

Practice Problem:

SM Explain the three major transmission impairments that degrade signals.

Solution:

Answer:

1. **Attenuation:** Loss of energy as signal propagates through resistance of the medium, measured in decibels (dB). Compensated with amplifiers.
2. **Distortion:** Change in waveform shape caused by different frequency components propagating at different speeds and suffering different attenuation.
3. **Noise:** Unwanted external energy induced onto the transmission line (thermal noise, induced noise from motors, crosstalk between wire pairs, impulse noise spikes).

10-Mark Questions (Long Answer / Comprehensive)

Practice Problem:

10M Describe the responsibilities and services of each layer of the OSI 7-layer reference model.

Solution:

Answer Summary:

Detail all 7 layers in top-down or bottom-up order as presented in Section 2:

1. **Physical Layer:** Transmission of raw bits, electrical/mechanical specifications, line coding.
2. **Data Link Layer:** Hop-to-hop framing, MAC addressing, flow control, CRC error control.
3. **Network Layer:** Source-to-destination routing, logical IP addressing, packetizing, fragmentation.
4. **Transport Layer:** Process-to-process delivery, port addressing, segmentation, end-to-end flow and error control.
5. **Session Layer:** Dialog control (half/full duplex), session synchronization and checkpointing.
6. **Presentation Layer:** Syntax translation, encryption/decryption, data compression.
7. **Application Layer:** User network interface (HTTP, DNS, FTP, email protocols).

Solved Questions from Past CIE & Model Papers

ARCHIVE

The following questions have been transcribed from the official Ramaiah CIE-1 test paper (Term 2025) and SEE archives in notes/cn/practice/cn-cie-1-and-2.pdf, solved with complete steps:

Ramaiah Internal Assessment - I (2025) · Q 1(a) [4 Marks · Blooms L3, CO1]

"We have a channel with a 2-MHz bandwidth. The SNR for this channel is 63. What are the appropriate bit rate and signal level? (To calculate signal level consider the bit rate to be 4Mbps less than the upper limit)."

Step 1: Compute Shannon upper limit:

$$C = B \log_2(1 + \text{extSNR}) = 2 \times 10^6 \times \log_2(1 + 63) = 2 \times 10^6 \times 6 = 12 \times 10^6 \text{ bps}$$

Step 2: Operating bit rate:

$$R = 12 \times 10^6 \text{ bps} - 4 \times 10^6 \text{ bps} = 8 \times 10^6 \text{ bps}$$

Step 3: Signal levels via Nyquist formula:

$$R = 2B \log_2(L) \implies 8 \times 10^6 = 2 \times (2 \times 10^6) \times \log_2(L) = 4 \times 10^6 \times \log_2(L)$$

$$\log_2(L) = 2 \implies L = 2^2 = 4 \text{ discrete signal levels.}$$

Ramaiah Internal Assessment - I (2025) · Q 2(a) [4 Marks · Blooms L4, CO1]

"Compare mesh topology with star topology. Draw a hybrid topology with a star backbone and three bus networks."

- **Mesh vs Star Comparison:** Mesh requires $n(n-1)/2$ dedicated physical links and $n-1$ I/O ports per device, ensuring zero traffic congestion and maximum privacy, but at immense cabling expense. Star requires only n links and 1 port per host connected to a central hub/switch, offering simpler installation, but suffers from a single point of failure at the hub.
- **Hybrid Topology Sketch:** See [Figure 1.2](#) above illustrating a central star switch linking three terminated bus segments (e.g. Finance, Engineering, Admin).

Ramaiah Internal Assessment - I (2025) · Q 2(b) [4 Marks · Blooms L3, CO1]
 "Convert the digital data 11001010 to digital signal by applying AMI and differential Manchester line coding schemes."

Given Bit Sequence: 1 1 0 0 1 0 1 0

- **AMI (Alternate Mark Inversion):** Binary 0 is represented by $0_{ext}V$. Binary 1 alternates between $+V$ and $-V$:
 Bit 1: $+V$ · Bit 1: $-V$ · Bit 0: $0V$ · Bit 0: $0V$ · Bit 1: $+V$ · Bit 0: $0V$
 · Bit 1: $-V$ · Bit 0: $0V$
- **Differential Manchester:** Always has a mid-bit transition for synchronization. Binary 0 causes an inversion at the start of the bit interval; binary 1 has no transition at the start.
 Bit trace: Assuming initial high state: Bit 1 (no transition at start, mid-bit drop to low); Bit 1 (no start transition, mid-bit rise to high); Bit 0 (inversion at start to low, mid-bit rise to high); Bit 0 (inversion at start to low, mid-bit rise to high); Bit 1 (no start transition, mid-bit drop to low); Bit 0 (inversion at start to high, mid-bit drop to low); Bit 1 (no start transition, mid-bit rise to high); Bit 0 (inversion at start to low, mid-bit rise to high).

Ramaiah Internal Assessment - I (2025) · Q 3(c) [7 Marks · Blooms L2, CO1]
 "What is transmission impairment? Briefly explain its causes with necessary diagrams."

Transmission impairment occurs when an analog or digital signal travels across an imperfect physical transmission medium, causing the received signal at the destination to differ from the injected signal at the source. The three primary causes are:

1. **Attenuation:** Loss of signal energy as heat due to conductor resistance; compensated via amplifiers ([Figure 1.3](#)).
2. **Distortion:** Occurs in composite signals because different frequency components propagate at different velocities (delay distortion), shifting arrival phases and altering waveform shape.
3. **Noise:** Spurious external electrical energy added to the signal (thermal noise, induced noise, crosstalk between adjacent wires, and impulse spikes).

Verified Practice Problems with Hidden Answers

▽ PRACTICE

Test your problem-solving speed on these 6 exam-pattern numericals. Click each card to reveal the complete step-by-step verified solution:

PRACTICE 1 · NYQUIST BIT RATE [3 MARKS]

Calculate the Nyquist maximum bit rate for a noiseless channel with bandwidth $B = 4\text{ kHz}$ using $L = 4$ signal levels.

 Practice Problem:

Show Step-by-Step Solution

Solution:

Formula: $C = 2B \log_2(L)$

$C = 2 \times 4000 \times \log_2(4) = 8000 \times 2 = 16,000 \text{ bps}$

Result: $C = 16 \text{ kbps}$

PRACTICE 2 · MESH TOPOLOGY LINKS [3 MARKS]

In a fully connected mesh network of 10 computers, how many physical cables and I/O ports per device are required?

 Practice Problem:

Show Step-by-Step Solution

Solution:

- Number of duplex links: $\frac{n(n-1)}{2} = \frac{10 \times 9}{2} = \mathbf{45 \text{ physical links}}$.

- I/O ports per computer: $n - 1 = 10 - 1 = \mathbf{9 \text{ ports}}$.

Result: 45 Links · 9 Ports per node

PRACTICE 3 · CASCADED AMPLIFIER & CABLE [4 MARKS]

A signal with $P_{in} = 50 \text{ mW}$ enters a cable with -6 dB loss, followed by an amplifier with $+12 \text{ dB}$ gain. Find the final output power P_{out} .

 Practice Problem:

Show Step-by-Step Solution

Solution:

Net gain: $dB_{total} = -6 \text{ dB} + 12 \text{ dB} = +6 \text{ dB}$.

$6 = 10 \log_{10} \left(\frac{P_{out}}{50} \right) \implies \frac{P_{out}}{50} = 10^{0.6} \approx 3.98107$

$$P_{out} = 50 \times 3.98107 = 199.05 \text{ mW}$$

Result: $P_{out} = 199.05 \text{ mW}$

PRACTICE 4 · SHANNON CHANNEL CAPACITY [5 MARKS]

Find the Shannon capacity of a telephone line with bandwidth $B = 3.1 \text{ kHz}$ and $SNR_{dB} = 35 \text{ dB}$.

 Practice Problem:

Show Step-by-Step Solution

Solution:

Linear $SNR = 10^{3.5} \approx 3162.28$.

$C = 3100 \times \log_2(1 + 3162.28) = 3100 \times 11.6272 =$
 $\mathbf{36,044.4 \text{ bps}} \approx \mathbf{36.04 \text{ kbps}}$

Result: $C = 36.04 \text{ kbps}$

PRACTICE 5 · BANDWIDTH ENGINEERING [5 MARKS]

What bandwidth B is required to transmit 56extkbps over a noiseless channel using 16-QAM ($L = 16$)?

 Practice Problem:

Show Step-by-Step Solution

Solution:

Bits per signal element: $r = \log_2(16) = 4\text{extbits}$.

$$C = 2B \log_2(L) \implies 56,000 = 2 \times B \times 4 = 8B$$

$$B = \frac{56,000}{8} = \mathbf{7,000\text{ ext{ Hz}}} = \mathbf{7\text{ ext{ kHz}}}$$

Result: Required Bandwidth $B = 7.000\text{ kHz}$

PRACTICE 6 · BANDWIDTH-DELAY PRODUCT [5 MARKS]

A 100extMbps link connects two routers across a 2000extkm optical fiber link ($v = 2 \times 10^8\text{extm/s}$). Compute the Bandwidth-Delay Product in bits and in Kilobytes.

 Practice Problem:

Show Step-by-Step Solution

Solution:

$$T_{\text{prop}} = \frac{2000 \times 10^3 \text{ ext{ m}}}{2 \times 10^8 \text{ ext{ m/s}}} = 0.01 \text{ ext{ s}} = 10 \text{ ext{ ms}}$$

$$\text{extBDP} = (100 \times 10^6 \text{ extbps}) \times 0.01 \text{ exts} = 1,000,000 \text{ ext{ BDP in Bytes}} = \frac{1,000,000}{8} = \mathbf{125,000 \text{ ext{ Bytes}}} = \mathbf{125 \text{ ext{ KB}}}$$

Result: BDP = 1,000,000 bits (125 KB in flight)